Scheduled Jumps at FDA Decisions: Derivations

1 Setup

Known decision time τ. For $t < \tau$,

$$\frac{\mathrm{d}S_t}{S_{t^-}} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t \qquad (\mathbb{P}), \tag{1}$$

$$\frac{\mathrm{d}S_t}{S_{t^-}} = r\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}} \qquad (\mathbb{Q}). \tag{2}$$

Single jump at τ:

$$S_\tau = S_{\tau^-}(1+J), \qquad J = \begin{cases} u > 0 & \text{w.p. } p\ (\mathbb{P}),\ q\ (\mathbb{Q}) \\ -d,\ d \in (0,1) & \text{w.p. } 1-p\ (\mathbb{P}),\ 1-q\ (\mathbb{Q}). \end{cases} \tag{3}$$

2 No-arbitrage at the jump

Over $[\tau^-, \tau]$ the diffusive drift is $O(\mathrm{d}t)$, so the martingale condition is

$$\mathbb{E}_{\mathbb{Q}}[\,1 + J \mid \mathcal{F}_{\tau^-}\,] = 1 \tag{4}$$

$$q(1+u_{\mathbb{Q}}) + (1-q)(1-d_{\mathbb{Q}}) = 1 \tag{5}$$

$$\Rightarrow \quad q\,u_{\mathbb{Q}} = (1-q)\,d_{\mathbb{Q}}. \tag{6}$$

3 Implied move and premium

ATM straddle, strike $K = S_{\tau^-}$, expiry τ^+. Payoff $= |S_\tau - K| = S_{\tau^-}|J|$. Price:

$$\text{Straddle}_{\tau^-} = e^{-r\,\mathrm{d}t}\,\mathbb{E}_{\mathbb{Q}}\big[\,S_{\tau^-}|J| \mid \mathcal{F}_{\tau^-}\big] \xrightarrow{\mathrm{d}t\to 0} S_{\tau^-}\,\mathbb{E}_{\mathbb{Q}}|J|, \tag{7}$$

$$\text{implied_move} \equiv \frac{\text{Straddle}_{\tau^-}}{S_{\tau^-}} = \mathbb{E}_{\mathbb{Q}}|J| = q\,u_{\mathbb{Q}} + (1-q)\,d_{\mathbb{Q}}, \tag{8}$$

$$\text{realized_move} = |J|, \tag{9}$$

$$\pi \equiv \mathbb{E}_{\mathbb{Q}}|J| - \mathbb{E}_{\mathbb{P}}|J|. \tag{10}$$

At finite maturity the straddle also carries diffusive time value $\theta_{\tau^-} \geq 0$, so $\text{Straddle}_{\tau^-}/S_{\tau^-} = \mathbb{E}_{\mathbb{Q}}|J| + \theta_{\tau^-}$ and the raw implied-move wedge $= \pi + \mathbb{E}[\theta_{\tau^-}] \geq \pi$ overstates the event premium; $\theta \to 0$ only as $\mathrm{d}t \to 0$. Seller P&L per unit spot, which nets the time value on buy-back:

$$\mathbb{E}_{\mathbb{P}}\left[\frac{\text{Straddle}_{\tau^-} - \text{Straddle}_{\tau^+}}{S_{\tau^-}}\right] = \mathbb{E}_{\mathbb{Q}}|J| - \mathbb{E}_{\mathbb{P}}|J| + o(1) = \pi + o(1). \tag{11}$$

4 Variance decomposition and identification

Proposition 1. *If the diffusive and jump log-increments are $\mathbb{Q}$-independent and σ^2 is continuous across τ,*

$$\mathrm{Var}_{\mathbb{Q}}\Big(\ln \tfrac{S_T}{S_0}\Big) = \sigma^2 T + V_J\,\mathbf{1}\{T \geq \tau\}, \quad V_J := \mathrm{Var}_{\mathbb{Q}}(\ln(1+J)), \tag{12}$$

and, with $\mathrm{MFIV}(T) = \mathrm{Var}_{\mathbb{Q}}(\ln S_T/S_0)$ *the model-free implied variance recovered from the strike continuum (Britten-Jones–Neuberger 2000; Carr–Madan 1998),*

$$V_J = \lim_{T\downarrow\tau} \mathrm{MFIV}(T) - \lim_{T\uparrow\tau} \mathrm{MFIV}(T). \tag{13}$$

Proof.

$$\ln \frac{S_T}{S_0} = \underbrace{\Big(r - \tfrac{\sigma^2}{2}\Big)T + \sigma W_T^{\mathbb{Q}}}_{X_{\mathrm{diff}}} + \mathbf{1}\{T \geq \tau\} \ln(1+J), \tag{14}$$

$$\mathrm{Var}_{\mathbb{Q}}\Big(\ln \tfrac{S_T}{S_0}\Big) = \mathrm{Var}_{\mathbb{Q}}(X_{\mathrm{diff}}) + \mathbf{1}\{T \geq \tau\}\mathrm{Var}_{\mathbb{Q}}(\ln(1+J)) \quad \text{(independence)} \tag{15}$$

$$= \sigma^2 T + \mathbf{1}\{T \geq \tau\}\, V_J. \tag{16}$$

At the money $\mathrm{IV}^2(T)\,T = \mathrm{Var}_{\mathbb{Q}}(\ln S_T/S_0) + o(1)$; $\sigma^2 T$ is continuous, so the one-sided limits differ by V_J. □

5 Downside asymmetry (price of crash insurance)

Events i: fair move $m_i = \mathbb{E}_{\mathbb{P}}|J_i|$, crash exposure $k_i \geq 0$, price of crash risk $\lambda \geq 0$:

$$c_i \equiv \mathbb{E}_{\mathbb{Q}}|J_i| = m_i + \lambda k_i, \qquad \pi_i = c_i - m_i = \lambda k_i, \qquad \mathbb{P}(\mathrm{rej} \mid i) = g(k_i),\ g \uparrow. \tag{17}$$

Proposition 2. *If $\lambda > 0$ and g is nondecreasing and nonconstant:*

$$\mathbb{E}[\pi_i] = \lambda\,\mathbb{E}[k_i] > 0, \qquad \mathbb{E}[\pi_i \mid \mathrm{rej}] \geq \mathbb{E}[\pi_i \mid \mathrm{app}]. \tag{18}$$

Proof. Prior $\varphi(k)$. Posteriors:

$$\varphi_R(k) = \frac{\varphi(k)g(k)}{\int \varphi g}, \qquad \varphi_A(k) = \frac{\varphi(k)(1-g(k))}{\int \varphi(1-g)}. \tag{19}$$

Likelihood ratio

$$\frac{\varphi_R(k)}{\varphi_A(k)} \propto \frac{g(k)}{1-g(k)} \ \uparrow \text{ in } k \tag{20}$$

is monotone (MLR) $\Rightarrow \varphi_R$ first-order stochastically dominates $\varphi_A \Rightarrow \mathbb{E}_R[k] \geq \mathbb{E}_A[k]$. Since $\pi_i = \lambda k_i$,

$$\mathbb{E}[\pi \mid \mathrm{rej}] = \lambda \mathbb{E}_R[k] \geq \lambda \mathbb{E}_A[k] = \mathbb{E}[\pi \mid \mathrm{app}]. \tag{21}$$

Unconditional: $\mathbb{E}[\pi] = \lambda \mathbb{E}[k]$ by iterated expectations. □

6 Implied approval probability: bounded martingale

$p_t \equiv \mathbb{Q}(\text{app} \mid \mathcal{F}_t) \in [0,1]$:

$$\mathrm{d}p_t = \nu_t \sqrt{p_t(1-p_t)}\,\mathrm{d}B_t, \qquad \mathcal{A} = \tfrac{1}{2}\nu_t^2\, p(1-p)\,\partial_{pp}, \qquad \mathbb{E}_{\mathbb{Q}}[\mathrm{d}p_t \mid \mathcal{F}_t] = 0. \tag{22}$$

Zero drift + boundedness $\Rightarrow p_t$ is a true $\mathbb{Q}$-martingale; $p_\tau \in \{0,1\}$. Pre-event:

$$\mathbb{E}_{\mathbb{Q}}[\text{payoff} \mid \mathcal{F}_t] \text{ is a martingale } \Rightarrow \text{ no predictable pre-}\tau \text{ drift.} \tag{23}$$

Breeden–Litzenberger (risk-neutral density and digital):

$$f_{\mathbb{Q}}(K) = e^{rT}\,\partial_{KK}C(K), \qquad \mathbb{Q}(S_T \geq K) = -e^{rT}\,\partial_K C(K), \tag{24}$$

and the implied approval probability is the digital at the approval-vs-rejection separating strike.

7 Optimal control

Per-event short-straddle return $\tilde{R}$, $\pi_s = \mathbb{E}_{\mathbb{P}}[\tilde{R}]$, $s^2 = \mathrm{Var}_{\mathbb{P}}(\tilde{R})$; wealth fraction f; $U(w) = w^{1-\gamma}/(1-\gamma)$.

$$V = \sup_{s,f}\ \mathbb{E}_{\mathbb{P}}\big[\,U(W_0(1+f\tilde{R}))\,\big]. \tag{25}$$

(i) Size — CRRA expansion

$$0 = \partial_f \mathbb{E}[U] = \mathbb{E}\big[\tilde{R}\,U'(W_0(1+f\tilde{R}))\big] = W_0^{-\gamma}\,\mathbb{E}\big[\tilde{R}\,(1+f\tilde{R})^{-\gamma}\big], \tag{26}$$

$$(1+f\tilde{R})^{-\gamma} \approx 1 - \gamma f\tilde{R}, \tag{27}$$

$$0 = \mathbb{E}[\tilde{R}] - \gamma f\,\mathbb{E}[\tilde{R}^2] \ \Rightarrow\ f^\star = \frac{\mathbb{E}[\tilde{R}]}{\gamma\,\mathbb{E}[\tilde{R}^2]} \approx \frac{\pi_s}{\gamma s^2}. \tag{28}$$

(i′) Size — exact log utility, two outcomes

Outcomes $g_+ > 0 > g_-$ w.p. $\rho, 1-\rho$; $\pi_s = \rho g_+ + (1-\rho)g_-$:

$$h(f) = \rho\ln(1+fg_+) + (1-\rho)\ln(1+fg_-), \tag{29}$$

$$h'(f) = \frac{\rho g_+}{1+fg_+} + \frac{(1-\rho)g_-}{1+fg_-} = 0, \tag{30}$$

$$0 = \rho g_+(1+fg_-) + (1-\rho)g_-(1+fg_+) \tag{31}$$

$$= \big[\rho g_+ + (1-\rho)g_-\big] + f g_+ g_-\big[\rho + (1-\rho)\big] \tag{32}$$

$$= \pi_s + f\,g_+g_- \ \Rightarrow\ f^\star = \frac{-\pi_s}{g_+g_-} = \frac{\pi_s}{-g_+g_-} > 0. \tag{33}$$

(ii) Break-even cost

Straddle $c = \mathbb{E}_{\mathbb{Q}}|J|$ per unit spot; round-trip spread κ:

$$\mathbb{E}_{\mathbb{P}}[\text{net}] = \pi_s - \kappa c > 0 \iff \kappa < \kappa^\star := \frac{\pi_s}{c}. \tag{34}$$

(iii) Entry timing

Pre-event carry $\Delta\Pi$ over $[s,\tau)$ with $\mathbb{E}_{\mathbb{P}}[\Delta\Pi]=0$, $\mathrm{Var}_{\mathbb{P}}(\Delta\Pi)=\beta(\tau-s)\uparrow$:

$$\mathrm{CE}(s)=\mathbb{E}[\Delta\Pi]-\tfrac{\gamma}{2}\mathrm{Var}(\Delta\Pi)=-\tfrac{\gamma}{2}\beta(\tau-s)\ \downarrow\ \text{in}\ (\tau-s) \tag{35}$$

$$\Rightarrow\ s^\star=\tau^-. \tag{36}$$

With predictable vega run-up $\mathbb{E}[\Delta\Pi]=v(\tau-s)$:

$$\mathrm{CE}(s)=v(\tau-s)-\tfrac{\gamma}{2}\beta(\tau-s)=\big(v-\tfrac{\gamma}{2}\beta\big)(\tau-s),\qquad \frac{\mathrm{d}\,\mathrm{CE}}{\mathrm{d}(\tau-s)}=v-\tfrac{\gamma}{2}\beta\ \Rightarrow\ \text{knife-edge}\ v=\tfrac{\gamma}{2}\beta. \tag{37}$$

8 Empirical values

$\mathbb{E}_{\mathbb{Q}}\lvert J\rvert=0.0726$	$\mathbb{E}_{\mathbb{P}}\lvert J\rvert=0.0338$	wedge $=0.0388$ $(t=3.37)$
$\pi_s=0.0129$ $(t=1.67)$	$c=0.0726$	$\kappa^\star=\pi_s/c=0.178$
app: $\mathbb{E}_{\mathbb{Q}}\lvert J\rvert=0.0546,\ \mathbb{E}_{\mathbb{P}}\lvert J\rvert=0.0303$	rej: $\mathbb{E}_{\mathbb{Q}}\lvert J\rvert=0.1025,\ \mathbb{E}_{\mathbb{P}}\lvert J\rvert=0.0395$	
$f^\star_{\log}=-\pi_s/(g_+g_-)=35.5$	$f^\star_{\gamma=1}=\pi_s/s^2=9.0$	$f^\star_{\gamma=3}=3.0.$